JIAZHI SUN

Email: jzsun2019@gmail.com | Phone: +86 130 7349 0960 | Website: ORCiD, Google Scholar

EDUCATION

Ph.D., Hohai University, China **Sep. 2020–Sep. 2026**

Thesis: Analytical Modeling of Groundwater Extraction in Coastal Aquifers under Complex Hydrogeological Conditions.
Advisor: Prof. Chunhui Lu

- Developed analytical models for coastal aquifers with sloping geometries, confined and unconfined conditions, low-permeability barriers, and spatial heterogeneity
- Used MODFLOW, MT3D, and SEAWAT through FloPy to simulate groundwater flow, solute transport, and freshwater–saltwater dynamics in coastal aquifers
- Evaluated how complex hydrogeological conditions influence seawater intrusion, freshwater–saltwater interface migration, and maximum safe pumping rates

Visiting Ph.D. Student, Ghent University, Belgium **Feb. 2025–Feb. 2026**

Host: Prof. Thomas Hermans

- Conducted hydrogeophysical field surveys and interpreted the data to characterize subsurface conditions
- Investigated the effects of aquifer heterogeneity and hydraulic connectivity on upper saline plume development and freshened submarine groundwater discharge

B.Eng., Hohai University, China **Sep. 2016–Jun. 2020**

Thesis: An analytical model for groundwater flow induced by an oscillatory pumping test based on lagging theory: A free surface equation considering the influence of the capillary fringe.
Advisor: Prof. Ching-Sheng Huang

- Derived analytical solutions for oscillatory pumping-induced flow, including capillary fringe effects, and applied finite element methods to simulate unconfined aquifer responses
- Gained practical training in hydrological measurements, GIS, flood forecasting, frequency analysis, streamflow computation, and reservoir regulation

PROFESSIONAL DEVELOPMENT

Seismic and CSEM Training School, Hannover, Germany **Oct. 06–08, 2025**

- Completed advanced training on geophysical methods for offshore freshened groundwater exploration
- Acquired hands-on skills in data processing, inversion, and interpretation of ERT and seismic datasets

Summer School on Machine Learning and Deep Learning, Nanjing, China **Aug. 15–20, 2022**

- Learned the fundamentals of machine learning, widely used algorithms, and their implementation in R
- Explored the basic concepts and history of deep learning, and current research frontiers

RESEARCH INTERESTS

Coastal hydrogeology, variable-density groundwater flow, freshwater–saltwater interactions, aquifer heterogeneity and uncertainty analysis, managed aquifer recharge and subsurface freshwater storage

AWARDS & HONORS

Competitive Selection, Seismic and CSEM Training School, Hannover, Germany	Oct. 2025
International Mobility Fellowship, Hohai University, China	2025–2026
First-Class Academic Scholarship, Hohai University, China	2020–2025

PUBLICATIONS

Peer-reviewed Journal Articles

1 | Zong, H., Wu, H., **Sun, J.**, Xu, W., & Lu, C. (**2026**). Analytical solutions for optimizing saltwater extraction and freshwater lens protection in saline riparian aquifers. ***Water Resources Research***, 62(9), e2026WR043972. https://doi.org/10.1029/2026WR043972

2 | Liu, Y., Lu, C., Ye, Y., **Sun, J.,** & Werner, A. D. (**2026**). Effect of alongshore aquifer property changes on the steady-state extent of seawater in coastal aquifers: analytical derivation and numerical validation. ***Water Resources Research***, 62(7), e2025WR043075. https://doi.org/10.1029/2025WR043075

3 | **Sun, J.,** Shi, W., Wu, H., Yang, J., & Lu, C. (**2026**). A novel analytical solution for quantifying seawater intrusion in coastal aquifers embedded with a cutoff wall. ***Advances in Water Resources,*** 105301. https://doi.org/10.1016/j.advwatres.2026.105301

4 | Xu, W., Lu, C., **Sun, J.,** Yang, X., & Werner, A. D. (**2026**). Analytical solutions for seawater intrusion from pumping and sea-level rise in coastal aquifers with a general-head inland boundary. ***Advances in Water Resources***, 105310. https://doi.org/10.1016/j.advwatres.2026.105310

5 | **Sun, J.,** Shi, W., Jing, M., & Lu, C. (**2025**). Quantifying the performance of the horizontal less permeable barrier in improving the maximum sustainable withdrawal rate in coastal aquifers. ***Advances in Water Resources***, 105000. https://doi.org/10.1016/j.advwatres.2025.105000

6 | **Sun, J.,** Wu, H., Yin, J., & Lu, C. (**2023**). Estimating the maximum safe pumping rate in sloping unconfined coastal aquifers. ***Water Resources Research***, 59(9), e2023WR034675. https://doi.org/10.1029/2023WR034675

Under review

7 | **Sun, J.,** Zhang, J., Shen, C., & Lu, C. Analytical quantification of maximum safe pumping rate in sloping confined coastal aquifers. (*Advances in Water Resources*)

8 | Zhang, J., Lu, C., **Sun, J.,** Hu, Y., Yan, M., & Li, L. Maximum sustainable pumping rate in coastal aquifers with landward horizontal heterogeneity: A two-zone semi-analytical solution. (*Water Resources Research, Major revision*)

9 | Zhang, J., Shi, W., **Sun, J.**, Yao, H., & Lu, C. Maximum pumping in head-controlled reclaimed coastal aquifers: effects of horizontal heterogeneity and areal recharge. (*Advances in Water Resources*)

10 | Wang, K., Kong, J., **Sun, J.,** Luo, Z., Wen, Y., & Zhang, F. An approximate analytical solution of the hillslope-storage Boussinesq model incorporating lateral unsaturated flow for rapid landslide risk assessment. (*Journal of Hydrology, Major revision*)

In preparation

11 | **Sun, J.,** Paepen, M., Vrancken, E., Zhang, J., Lu, C., & Hermans, T. Impact of heterogeneity and hydraulic

connectivity on upper saline plume and freshened submarine groundwater discharge topology.

12 | **Sun, J.,** & Lu, C. A meshless model for seawater intrusion in heterogeneous coastal aquifers with complex boundary conditions.

13 | Yang, X., **Sun, J.**, Huang, X., Lu, C., & Luo, J. A two-dimensional analytical model of transient vapor intrusion.

Conference

1 | **Sun, J.,** Wu, H., Yin, J., and Lu, C.: Analytical Estimation of Maximum Safe Pumping Rate in Sloping Confined and Unconfined Coastal Aquifers, **EGU General Assembly 2025***,* Vienna, Austria, 27 Apr–02 May 2025, EGU25-8126, https://doi.org/10.5194/egusphere-egu25-8126. (**Oral**)

2 | Zhang, J., Lu, C., and **Sun, J.**: Finite Land Reclamation at Coastal Region: Impact of Heterogeneity on Groundwater and Seawater Extent, **EGU General Assembly 2025***,* Vienna, Austria, 27 Apr–2 May 2025, EGU25-7996, https://doi.org/10.5194/egusphere-egu25-7996

3 | **Sun, J.,** Fan, B., and Lu, C.: Effects of Artificially Reducing Hydraulic Conductivity of Coastal Aquifers on Maximum Sustainable Withdrawal Rate, **EGU General Assembly 2024***,* Vienna, Austria, 14–19 Apr. 2024, EGU24-2205, https://doi.org/10.5194/egusphere-egu24-2205. (**Poster**)

4 | Xu, W., Fan, B., **Sun, J.,** Lu, C.: Analytical assessment of available freshwater resources in coastal aquifers with a general-head inland boundary amidst sea level rise, **AOGS 21st annual meeting***,* PyeongChang, South Korea, 23–28 Jun. 2024, AOGS2024-23752

Patent

1 | Yin, J., Huang, Y., Lu, C., **Sun, J.**, et al., An alternative modeling approach for predicting saltwater intrusion dynamics under uncertainty conditions. **Patent No. ZL202410259227.1.** (In Chinese)

2 | Lu, C., **Sun, J.**, An analytical method for calculating the effectiveness of cutoff walls in preventing seawater intrusion. (Pending, 2023, In Chinese)

JOURNAL REVIEW

Water Resources Research, Journal of Hydrology, Watershed Ecology and the Environment, Advances in Water Resources, Water Research, Science of the Total Environment, Hydrogeology Journal

SKILLS

Groundwater and Solute Transport Modeling

- Derived analytical models for seawater intrusion in coastal areas and island aquifers, vapor intrusion, sloping aquifers, and hillslope storage under varying boundary conditions
- Proficient in using MODFLOW, MT3D, and SEAWAT (FloPy), and COMSOL for simulating groundwater flow, solute transport, seawater intrusion, and denitrification in coastal aquifers

Experimental Skills

- Experienced in laboratory-scale experiments on seawater intrusion, freshwater lens dynamics, offshore freshened groundwater formation, and solute transport within coastal aquifers

Programming and Numerical Tools

Python, Mathematica, and MATLAB

Languages

Mandarin (Native) and English (C1, IELTS score: 7.5)

Professional Memberships

European Geosciences Union (EGU) and International Association of Hydrogeologists (IAH)

MENTORING EXPERIENCE

Supervised undergraduate research under the Hohai University Innovation Training Program

1 | Xie, W., Formation and evolution mechanisms of offshore groundwater, 2023–2024

2 | Zhao, Y., Groundwater fluctuations in coastal unconfined aquifers influenced by managed aquifer recharge, 2024–2025

3 | Wang, Z., Estimating the maximum pumping rate of horizontal wells in coastal areas, 2025–2026

4 | Sun, Y., Analytical and numerical modeling of vadose zone recharge for mitigating seawater intrusion, 2026–2027 (Ongoing)

REFERENCES

Prof. Chunhui Lu

Director, State Key Laboratory of Water Disaster Prevention, Hohai University, China
Phone: +86 13770838570 | Email: clu@hhu.edu.cn

Prof. Thomas Hermans

Department of Geology, Ghent University, Belgium
Phone: +32 0499138853 | Email: Thomas.Hermans@UGent.be

Prof. Chong Sheng

College of Hydrology and Water Resources, Hohai University, China
Phone: +86 18225542849 | Email: csheng@hhu.edu.cn

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